

Crypto Market Analysis - 2025-07-14

Market Summary

The following table summarizes the market's phase based on a multi-factor model including Price, Open Interest, and CVD.

Asset	Market Phase	Days in Regime	Liquidations	Sharpe (20d)	Sortino (20d)	CVaR (20d)
BTCUSDT	Neutral	3 days	Normal	8.70	12.28	-1.46%
ETHUSDT	Neutral	2 days	Normal	7.46	11.27	-3.24%
SOLUSDT	Neutral	13 days	Normal	4.46	4.94	-5.12%
BNBUSDT	Neutral	13 days	Normal	7.48	7.79	-1.61%
XRPUSDT	Bullish	4 days	Normal	10.91	14.86	-3.68%
ADAUSDT	Bullish	6 days	Normal	6.42	7.73	-5.28%
TRXUSDT	Neutral	14 days	Normal	9.18	11.10	-1.50%

Market Overview

Overall market sentiment remains cautiously neutral, with compression signals dominating in five of seven major crypto pairs and only 29% displaying bullish trends. Risk-adjusted returns are generally healthy—Bitcoin’s Sharpe ratio sits at 8.7, Ethereum and BNB both at 7.5, and TRX at an impressive 9.2—while XRP leads the pack at 10.9 even as its bullish momentum sets it apart from the broader consolidation. This mixed backdrop suggests investors are digesting recent gains and weighing near-term catalysts before committing to a sustained directional move.

Viewed through the lens of Bitcoin’s four-year cycle, we are entering the latter innings of the post-halving bull phase, which historically peaks roughly 12–18 months after the event. The prevailing compression across most large-caps is consistent with a late-cycle plateau, as markets wrestle with profit-taking, macro headwinds, and positioning for the next major impulse. While healthy Sharpe ratios point to resilient risk appetite, institutional players should remain vigilant for a potential shift toward either a renewed breakout or a corrective reset as we approach the cycle’s high-water mark.

BTCUSDT Analysis

Current Signal: Compression

Prevailing Regime: The market has been in a **Neutral** state for **3 consecutive days**.

Latest Market Values:

Price (Intraday)	Open Interest	CVD
\$122,578.01 (2025-07-14 09:00 UTC)	\$10.33B	1.57k coins

Asset Summary:

- The market is in a state of extremely low realized volatility. This indicates a 'compression' regime where the market is coiling and building energy.

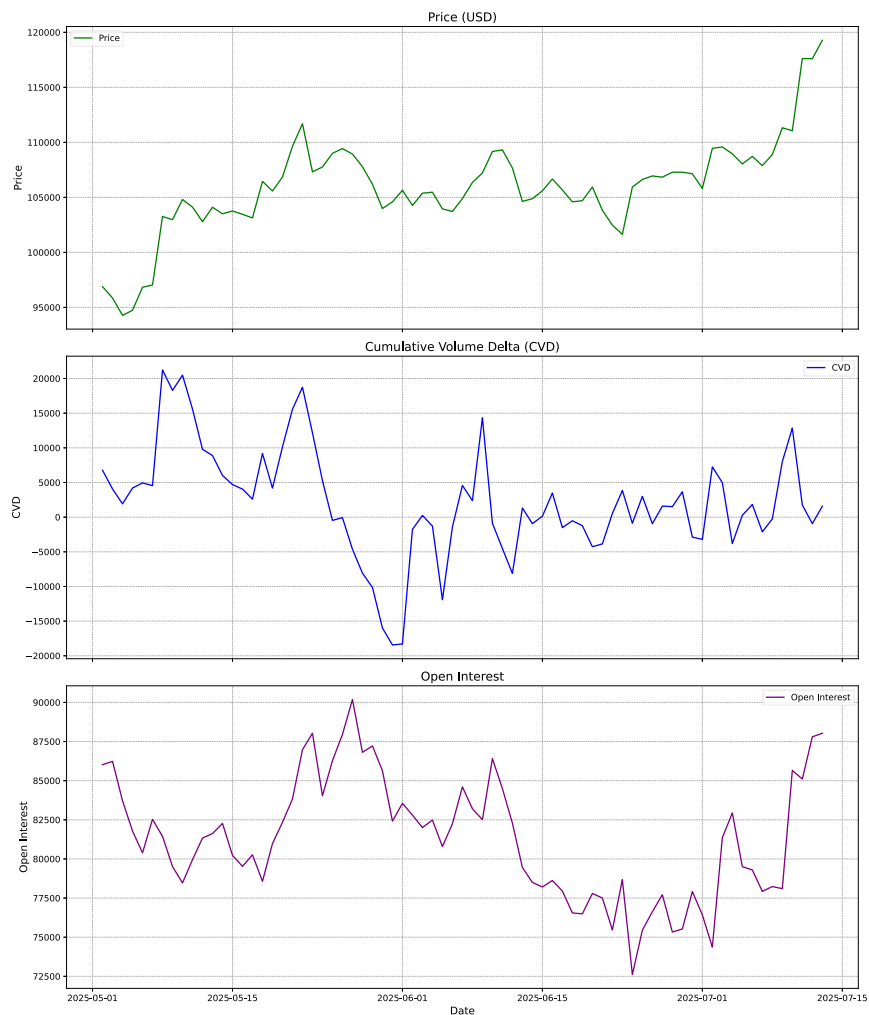
What to Watch For:

- Directional trades are low probability. Wait for volatility to expand, signaling the start of a new, more reliable move.

Longer-Term Observations:

- False Breakout Detected (2025-07-10):** The market's attempt at an **upward** breakout from compression failed to find follow-through, quickly returning to a low-volatility state. This rejection suggests a lack of conviction and reinforces the current range-bound thesis.

Daily Metrics Overview - BTCUSDT



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The market is in a quiet, neutral state with no unusual liquidation pressure.

Risk Metrics:

- **Volatility State:** Normal
- **RSI:** Overbought conditions
- **Sharpe Ratio (20d):** Excellent risk-adjusted returns
- **Sortino Ratio (20d):** Strong downside-adjusted returns
- **CVaR (20d):** Low tail risk
- **Omega Ratio (20d):** Strong gain/loss ratio
- **Max Drawdown (20d):** Minor recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	111,233.99	Compression	UP	UP	UP
2025-07-10	116,010.00	Bullish Trending	UP	UP	UP
2025-07-11	117,527.66	Compression	UP	UP	DOWN
2025-07-12	117,420.00	Compression	UP	UP	DOWN
2025-07-13	119,086.64	Compression	UP	UP	UP

Daily Technical Analysis for BTCUSDT (180 periods)



Bitcoin’s ongoing compression reflects a standoff that often precedes a sharp directional move once volume ramps up. In the next one to two weeks, a sustained push above \$125,000 would likely spark fresh upside momentum, whereas a drop under \$118,000 could usher in a more pronounced pullback. Keep a close eye on trading volume around these thresholds—its expansion will be the key signal confirming any decisive breakout or breakdown.

Technical Interpretation

The current 8-state model signal is **Compression**. The asset is in a **strong uptrend**, with the current price (\$121,026.29) above both the 50-day SMA (\$107,654.54) and the 200-day SMA (\$97,129.26). Volatility is elevated, suggesting active and trending market conditions. RSI is at **81.4**, indicating overbought conditions. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

AI Technical Perspective

The market’s strong advance above both SMAs with expanding Bollinger Bands confirms robust bullish momentum, but the overextended RSI warns of short-term fatigue. Watching for bearish RSI divergence or a close back inside the bands will be crucial for signaling the next leg. Traders should consider locking in profits with a tight trailing stop or waiting for a pullback toward the mid-band or SMA for a better entry.

Market State

Metric	Value
Daily Return (20d Avg)	1.63%
Volatility State	Normal
RSI	81.4 (Overbought)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	8.70
Sortino Ratio	
- 20-day	12.28
- 60-day	3.25

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-1.46%	-3.23%	-
Omega Ratio	3.55	1.53	-
Max Drawdown	-2.47%	-9.61%	-12.77%

ETHUSDT Analysis

Current Signal: Compression

Prevailing Regime: The market has been in a **Neutral** state for **2 consecutive days**.

Latest Market Values:

Price (Intraday)	Open Interest	CVD
\$3,039.77 (2025-07-14 09:00 UTC)	\$5.79B	-573.62 coins

Asset Summary:

- The market is in a state of extremely low realized volatility. This indicates a 'compression' regime where the market is coiling and building energy.

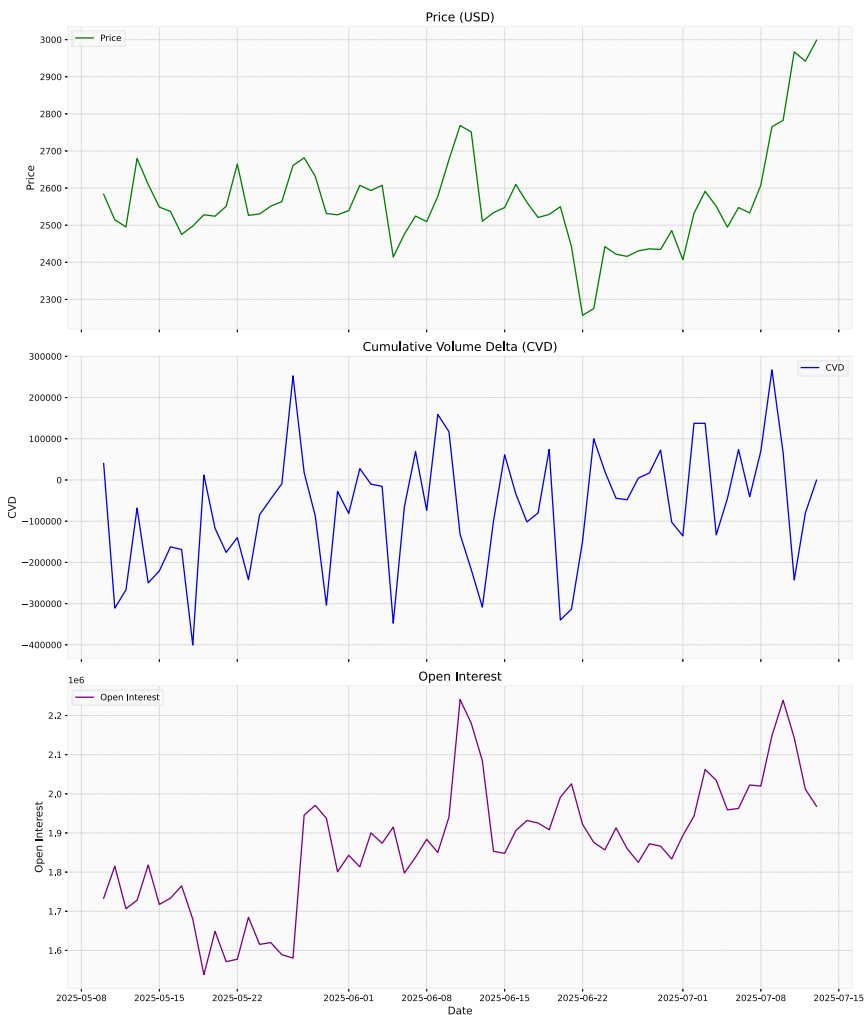
What to Watch For:

- Directional trades are low probability. Wait for volatility to expand, signaling the start of a new, more reliable move.

Longer-Term Observations:

- False Breakout Detected (2025-07-09):** The market's attempt at an **upward** breakout from compression failed to find follow-through, quickly returning to a low-volatility state. This rejection suggests a lack of conviction and reinforces the current range-bound thesis.

Daily Metrics Overview - ETHUSDT



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The market is in a quiet, neutral state with no unusual liquidation pressure.

Risk Metrics:

- **Volatility State:** Normal
- **RSI:** Overbought conditions
- **Sharpe Ratio (20d):** Excellent risk-adjusted returns
- **Sortino Ratio (20d):** Strong downside-adjusted returns
- **CVaR (20d):** Moderate tail risk
- **Omega Ratio (20d):** Strong gain/loss ratio
- **Max Drawdown (20d):** Minor recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	2,768.74	Bullish Trending	UP	UP	UP
2025-07-10	2,951.29	Bullish Exhaustion	UP	UP	DOWN
2025-07-11	2,958.22	Bullish Exhaustion	UP	UP	DOWN
2025-07-12	2,943.28	Compression	UP	DOWN	UP
2025-07-13	2,972.03	Compression	UP	DOWN	UP

Daily Technical Analysis for ETHUSDT (180 periods)



Technical Interpretation

The current 8-state model signal is **Compression**. The asset is in a **strong uptrend**, with the current price (\$3,033.12) above both the 50-day SMA (\$2,579.04) and the 200-day SMA (\$2,474.90). Volatility is elevated, suggesting active and trending market conditions. RSI is at **78.5**, indicating overbought conditions. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

Market State

Metric	Value
Daily Return (20d Avg)	2.06%
Volatility State	Normal
RSI	78.5 (Overbought)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	7.46
Sortino Ratio	
- 20-day	11.27
- 60-day	2.17

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-3.24%	-5.76%	-
Omega Ratio	3.24	1.36	-
Max Drawdown	-3.80%	-20.90%	-29.20%

SOLUSDT Analysis

Current Signal: Compression

Prevailing Regime: The market has been in a **Neutral** state for **13 consecutive days**.

Latest Market Values:

Price (Intraday)	Open Interest	CVD
\$167.46 (2025-07-14 09:00 UTC)	\$1.46B	-136.76k coins

Asset Summary:

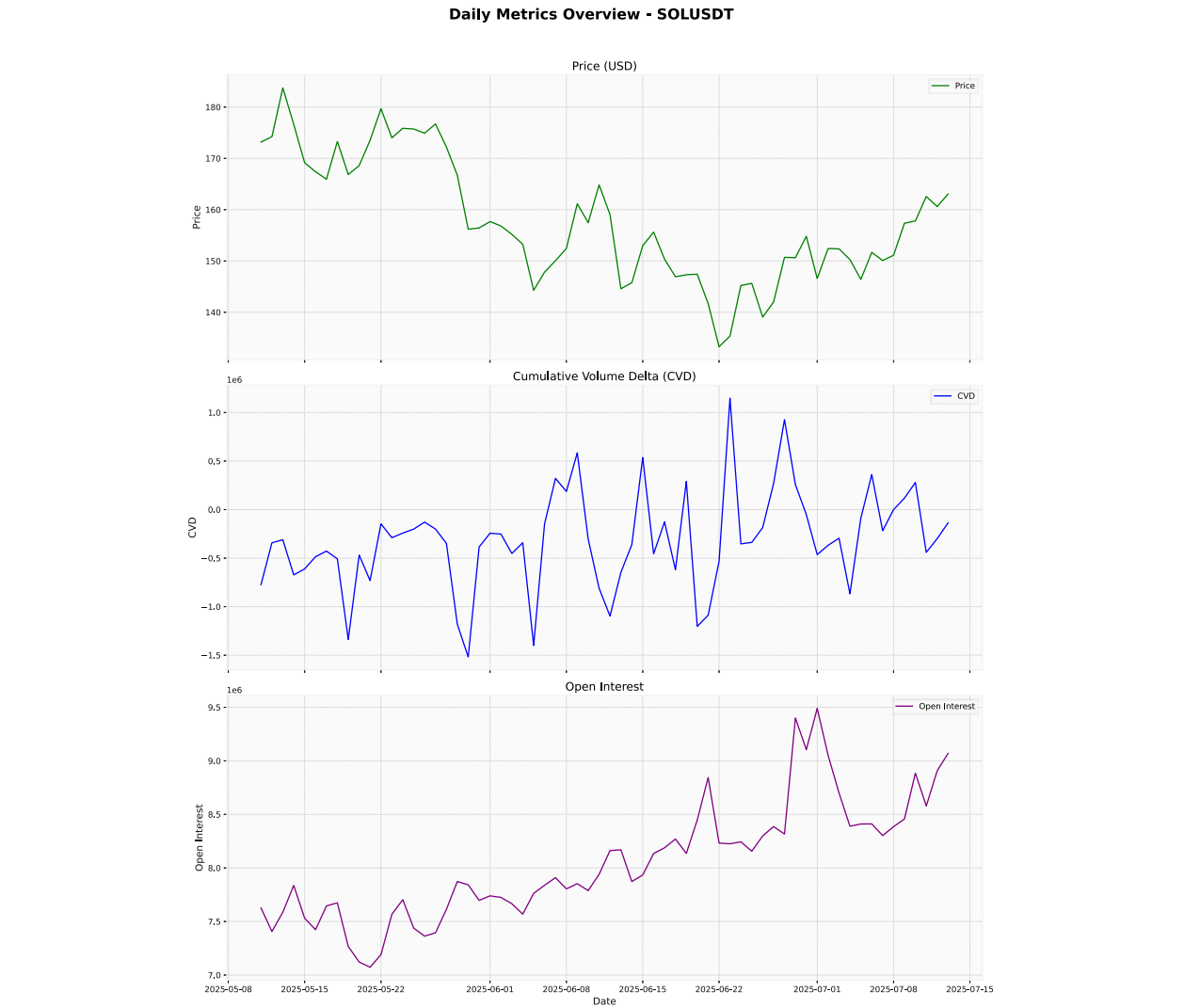
- The market is in a state of extremely low realized volatility. This indicates a 'compression' regime where the market is coiling and building energy.

What to Watch For:

- Directional trades are low probability. Wait for volatility to expand, signaling the start of a new, more reliable move.

Longer-Term Observations:

- False Breakout Detected (2025-07-07):** The market's attempt at an uncertain breakout from compression failed to find follow-through, quickly returning to a low-volatility state. This rejection suggests a lack of conviction and reinforces the current range-bound thesis.
- Stealth Distribution (since 2025-07-07):** While the price has remained in a consolidation range, the Cumulative Volume Delta (CVD) has been steadily **falling**. This suggests quiet distribution may be underway.



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The market is in a quiet, neutral state with no unusual liquidation pressure.

Risk Metrics:

- **Volatility State:** Normal
- **RSI:** Neutral conditions
- **Sharpe Ratio (20d):** Excellent risk-adjusted returns
- **Sortino Ratio (20d):** Strong downside-adjusted returns
- **CVaR (20d):** High tail risk
- **Omega Ratio (20d):** Moderate gain/loss ratio
- **Max Drawdown (20d):** Moderate recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	157.05	Compression	UP	UP	UP
2025-07-10	164.34	Compression	UP	UP	UP
2025-07-11	162.93	Compression	UP	UP	DOWN

2025-07-12	160.69	Compression	UP	UP	DOWN
2025-07-13	161.15	Compression	UP	UP	UP

Daily Technical Analysis for SOLUSDT (180 periods)



Technical Interpretation

The current 8-state model signal is **Compression**. The asset is in a long-term downtrend (50-day SMA is below 200-day SMA), but is currently experiencing a **relief rally** with the price (\$166.15) above the 50-day SMA (\$153.02). Volatility is **extremely low** (compression percentile < 30%), suggesting the market is coiling and building energy for a potentially sharp move. RSI is neutral at **61.3**. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

Market State

Metric	Value
Daily Return (20d Avg)	3.10%
Volatility State	Normal
RSI	61.3 (Neutral)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	4.46
Sortino Ratio	
- 20-day	4.94

- 60-day	0.05
----------	------

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-5.12%	-5.84%	-
Omega Ratio	1.80	1.02	-
Max Drawdown	-5.12%	-26.70%	-28.32%

--

BNBUSDT Analysis

Current Signal: Compression

Prevailing Regime: The market has been in a **Neutral** state for **13 consecutive days**.

Latest Market Values:

Price (Intraday)	Open Interest	CVD
\$704.53 (2025-07-14 09:00 UTC)	\$338.00M	-8.06k coins

Asset Summary:

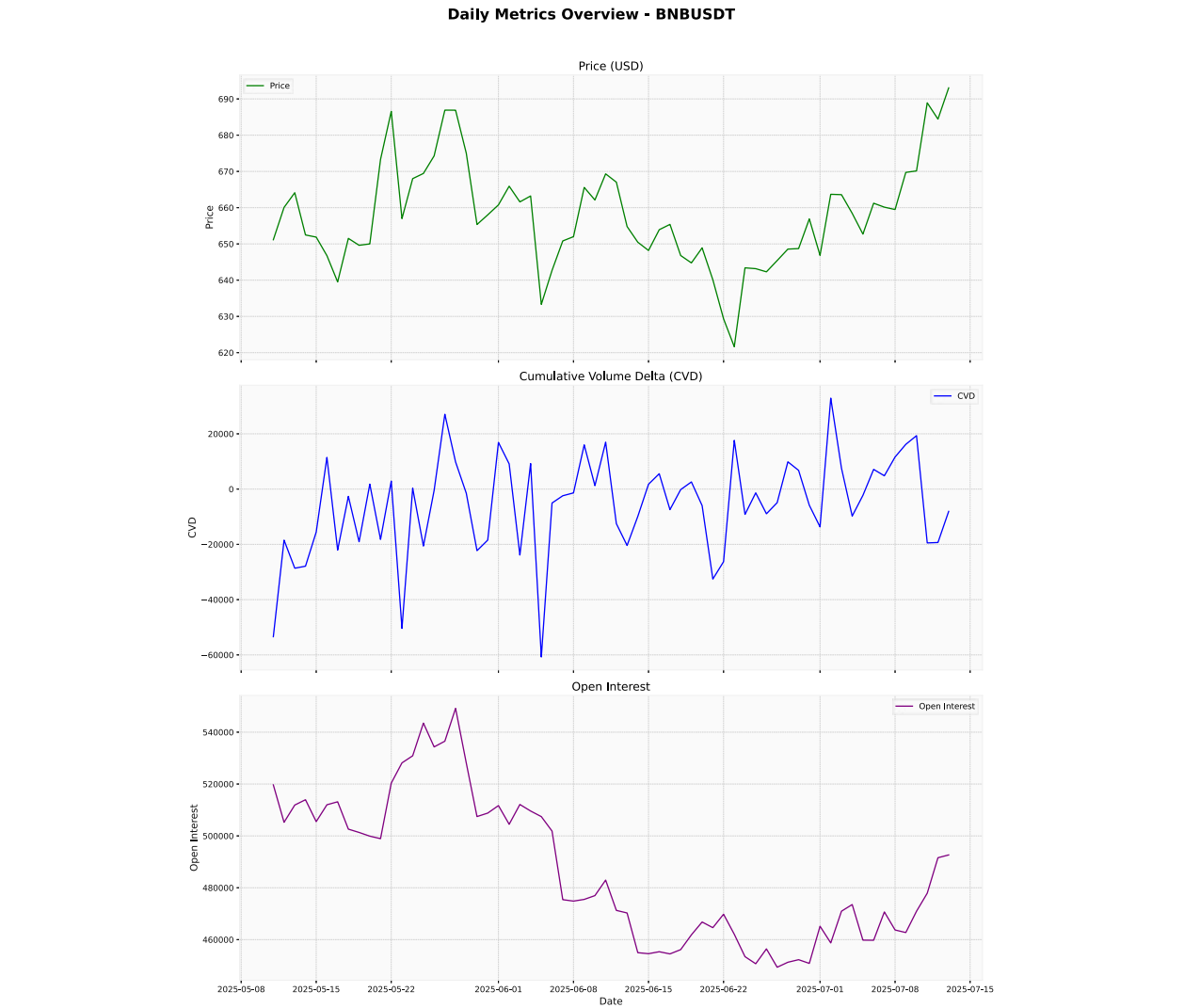
- The market is in a state of extremely low realized volatility. This indicates a 'compression' regime where the market is coiling and building energy.

What to Watch For:

- Directional trades are low probability. Wait for volatility to expand, signaling the start of a new, more reliable move.

Longer-Term Observations:

- False Breakout Detected (2025-06-29):** The market's attempt at an **upward** breakout from compression failed to find follow-through, quickly returning to a low-volatility state. This rejection suggests a lack of conviction and reinforces the current range-bound thesis.
- Stealth Distribution (since 2025-07-07):** While the price has remained in a consolidation range, the Cumulative Volume Delta (CVD) has been steadily **falling**. This suggests quiet distribution may be underway.



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The market is in a quiet, neutral state with no unusual liquidation pressure.

- Risk Metrics:**
- Volatility State:** Normal
 - RSI:** Overbought conditions
 - Sharpe Ratio (20d):** Excellent risk-adjusted returns
 - Sortino Ratio (20d):** Strong downside-adjusted returns
 - CVaR (20d):** Low tail risk
 - Omega Ratio (20d):** Strong gain/loss ratio
 - Max Drawdown (20d):** Minor recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	668.68	Compression	UP	DOWN	UP
2025-07-10	685.80	Compression	UP	UP	UP
2025-07-11	689.99	Compression	UP	UP	DOWN

2025-07-12	685.86	Compression	UP	UP	DOWN
2025-07-13	692.00	Compression	UP	UP	UP

Daily Technical Analysis for BNBUSDT (180 periods)



Technical Interpretation

The current 8-state model signal is **Compression**. The asset is in a **strong uptrend**, with the current price (\$698.20) above both the 50-day SMA (\$658.30) and the 200-day SMA (\$640.48). Volatility is lower than average, indicating a period of consolidation. RSI is at **71.6**, indicating overbought conditions. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

Market State

Metric	Value
Daily Return (20d Avg)	0.90%
Volatility State	Normal
RSI	71.6 (Overbought)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	7.48
Sortino Ratio	
- 20-day	7.79

- 60-day	1.23
----------	------

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-1.61%	-3.91%	-
Omega Ratio	2.77	1.25	-
Max Drawdown	-1.61%	-10.35%	-13.12%

XRPUSD Analysis

Current Signal: **Bullish** Trending

Prevailing Regime: The market has been in a **Bullish** state for **4 consecutive days**.

Latest Market Values:

Price (Intraday)	Open Interest	CVD
\$2.98 (2025-07-14 09:00 UTC)	\$1.01B	-20.09M coins

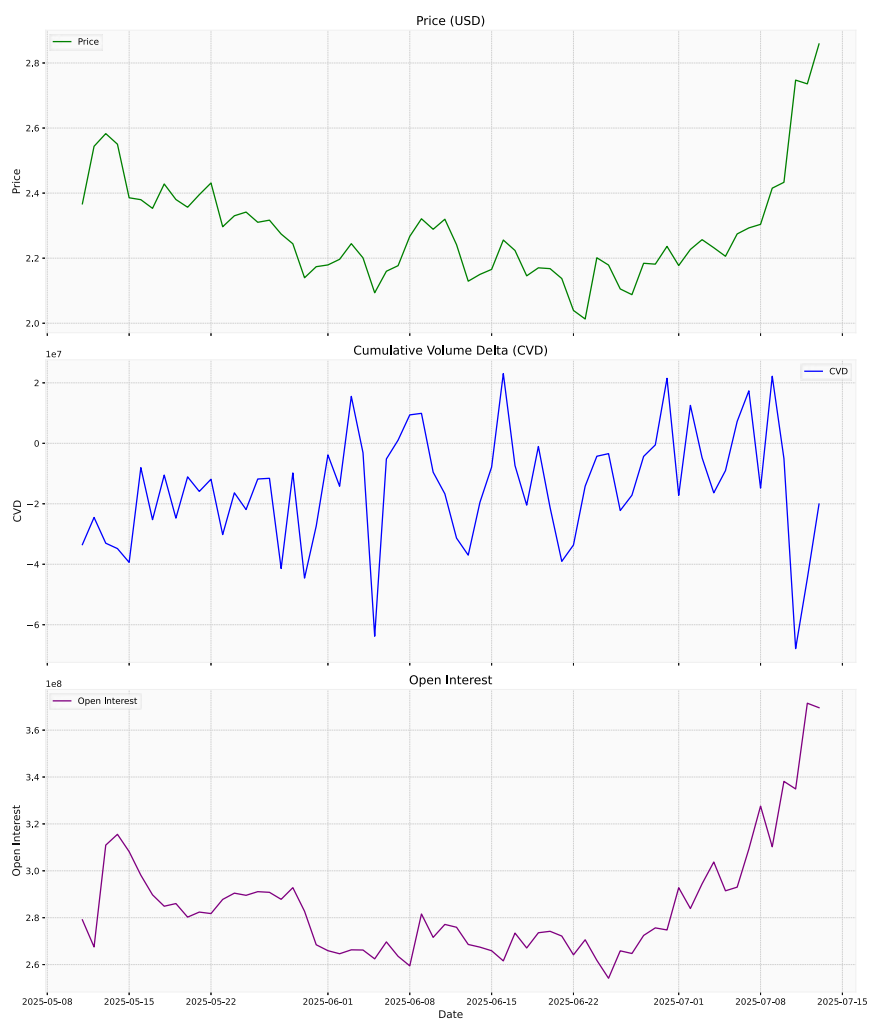
Asset Summary:

- Price, Open Interest, and CVD are all signaling a strong **upward** trend. This is a high-conviction **bullish** signal suggesting broad market participation.

What to Watch For:

- Bullish** continuation is likely. Watch for OI or CVD to falter as an early warning sign of weakness.

Daily Metrics Overview - XRPUSD



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The **bullish** trend is proceeding with normal levels of liquidations, suggesting stable and organic price action.

Risk Metrics:

- **Volatility State:** Normal
- **RSI:** Overbought conditions
- **Sharpe Ratio (20d):** Excellent risk-adjusted returns
- **Sortino Ratio (20d):** Strong downside-adjusted returns
- **CVaR (20d):** Moderate tail risk
- **Omega Ratio (20d):** Strong gain/loss ratio
- **Max Drawdown (20d):** Minor recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	2.40	Compression	UP	DOWN	UP
2025-07-10	2.55	Bullish Exhaustion	UP	UP	DOWN
2025-07-11	2.73	Bullish Exhaustion	UP	UP	DOWN
2025-07-12	2.74	Bullish Exhaustion	UP	UP	DOWN
2025-07-13	2.84	Bullish Trending	UP	UP	UP

Daily Technical Analysis for XRPUSDT (180 periods)



Technical Interpretation

The current 8-state model signal is **Bullish Trending**. The asset is in a long-term downtrend (50-day SMA is below 200-day SMA), but is currently experiencing a **relief rally** with the price (\$2.98) above the 50-day SMA (\$2.26). Volatility is elevated, suggesting active and trending market conditions. RSI is at **89.1**, indicating

overbought conditions. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

Market State

Metric	Value
Daily Return (20d Avg)	5.13%
Volatility State	Normal
RSI	89.1 (Overbought)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	10.91
Sortino Ratio	
- 20-day	14.86
- 60-day	3.04

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-3.68%	-5.02%	-
Omega Ratio	4.76	1.46	-
Max Drawdown	-3.68%	-17.03%	-29.47%

ADAUSDT Analysis

Current Signal: Bullish Trending

Prevailing Regime: The market has been in a Bullish state for **6 consecutive days**.

Latest Market Values:

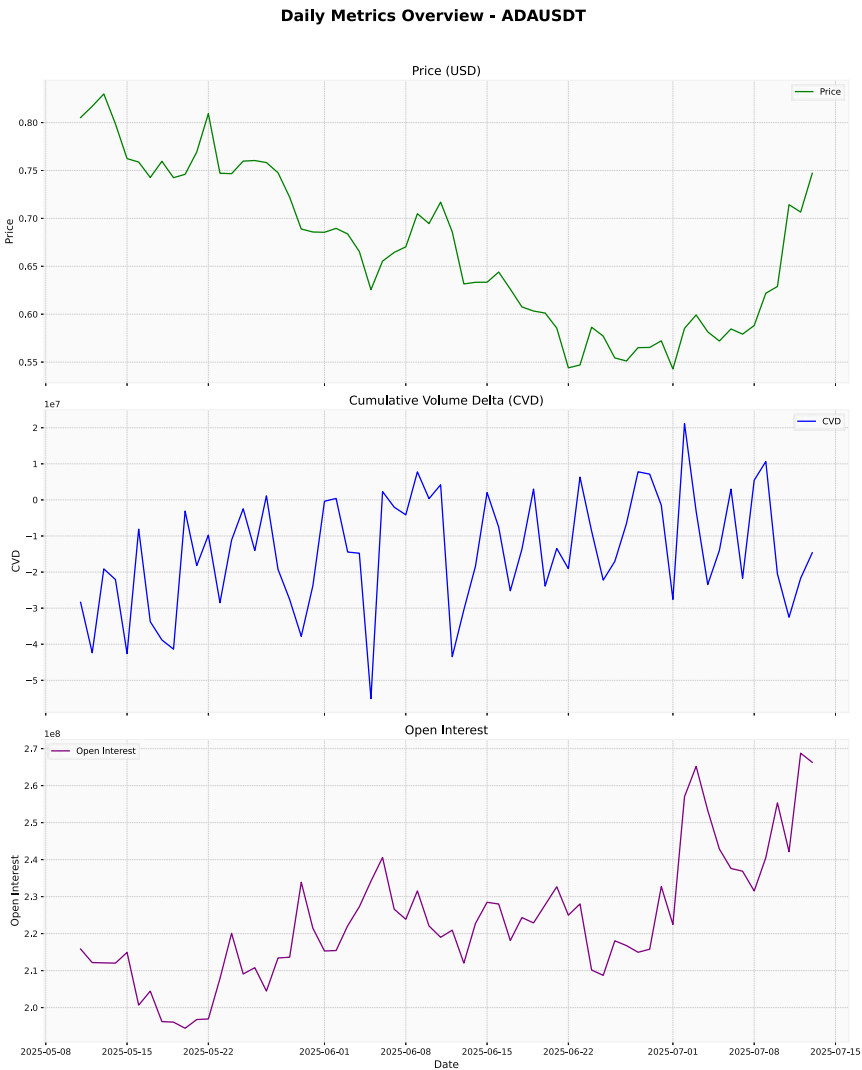
Price (Intraday)	Open Interest	CVD
\$0.75 (2025-07-14 09:00 UTC)	\$188.73M	-14.64M coins

Asset Summary:

- Price, Open Interest, and CVD are all signaling a strong upward trend. This is a high-conviction bullish signal suggesting broad market participation.

What to Watch For:

- Bullish continuation is likely. Watch for OI or CVD to falter as an early warning sign of weakness.



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The bullish trend is proceeding with normal levels of liquidations, suggesting stable and organic price action.

Risk Metrics:

- **Volatility State:** Normal
- **RSI:** Overbought conditions
- **Sharpe Ratio (20d):** Excellent risk-adjusted returns
- **Sortino Ratio (20d):** Strong downside-adjusted returns
- **CVaR (20d):** High tail risk
- **Omega Ratio (20d):** Strong gain/loss ratio
- **Max Drawdown (20d):** Moderate recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	0.62	Bullish Trending	UP	UP	UP
2025-07-10	0.68	Bullish Exhaustion	UP	UP	DOWN
2025-07-11	0.71	Bullish Exhaustion	UP	UP	DOWN
2025-07-12	0.71	Bullish Trending	UP	UP	UP
2025-07-13	0.74	Bullish Trending	UP	UP	UP

Daily Technical Analysis for ADAUSDT (180 periods)



Technical Interpretation

The current 8-state model signal is **Bullish Trending**. The asset is in a long-term downtrend (50-day SMA is below 200-day SMA), but is currently experiencing a **relief rally** with the price (\$0.75) above the 50-day SMA (\$0.64). Volatility is elevated, suggesting active and trending market conditions. RSI is at **79.0**, indicating

overbought conditions. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

Market State

Metric	Value
Daily Return (20d Avg)	1.25%
Volatility State	Normal
RSI	79.0 (Overbought)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	6.42
Sortino Ratio	
- 20-day	7.73
- 60-day	0.05

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-5.28%	-6.31%	-
Omega Ratio	2.46	1.02	-
Max Drawdown	-6.11%	-33.04%	-35.53%

TRXUSDT Analysis

Current Signal: Compression

Prevailing Regime: The market has been in a **Neutral** state for **14 consecutive days**.

Latest Market Values:

Price (Intraday)	Open Interest	CVD
\$0.30 (2025-07-14 09:00 UTC)	\$128.24M	3.92M coins

Asset Summary:

- The market is in a state of extremely low realized volatility. This indicates a 'compression' regime where the market is coiling and building energy.

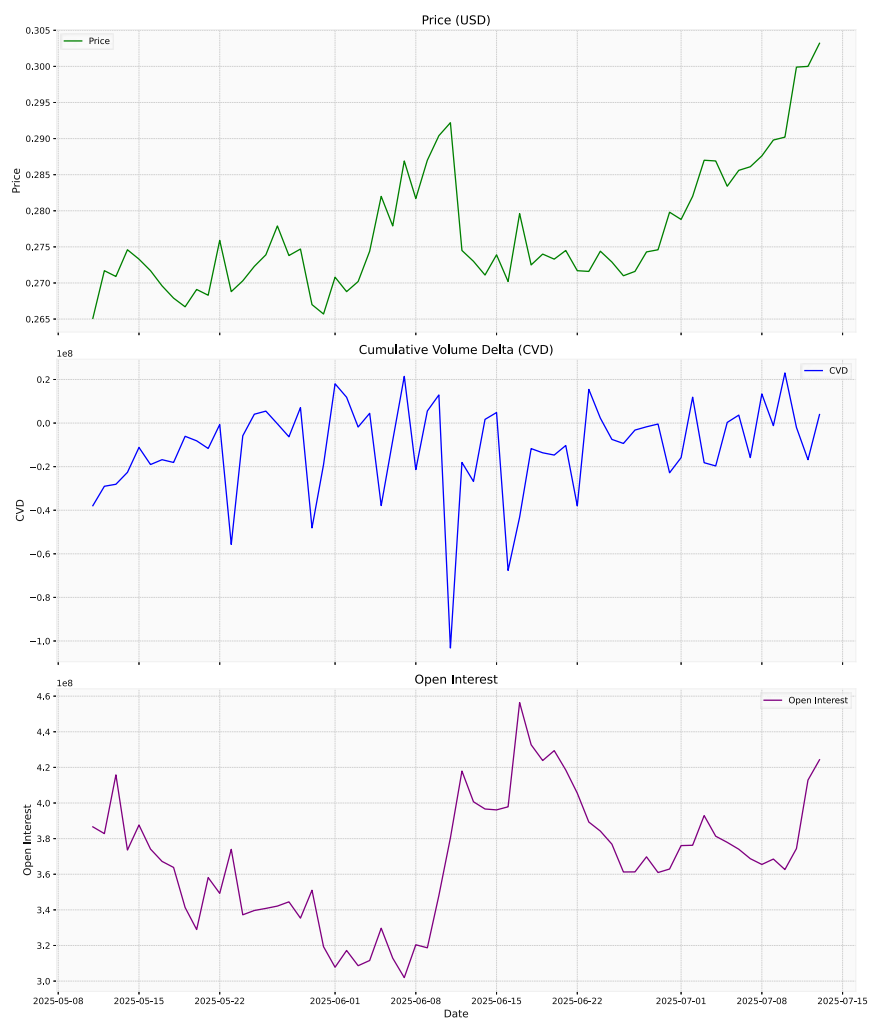
What to Watch For:

- Directional trades are low probability. Wait for volatility to expand, signaling the start of a new, more reliable move.

Longer-Term Observations:

- False Breakout Detected (2025-06-26):** The market's attempt at an **downward** breakout from compression failed to find follow-through, quickly returning to a low-volatility state. This rejection suggests a lack of conviction and reinforces the current range-bound thesis.

Daily Metrics Overview - TRXUSDT



Instability Index:

- The current instability level is **Normal**. This index measures the risk of liquidations in the market. The market is in a quiet, neutral state with no unusual liquidation pressure.

Risk Metrics:

- **Volatility State:** Low
- **RSI:** Overbought conditions
- **Sharpe Ratio (20d):** Excellent risk-adjusted returns
- **Sortino Ratio (20d):** Strong downside-adjusted returns
- **CVaR (20d):** Low tail risk
- **Omega Ratio (20d):** Strong gain/loss ratio
- **Max Drawdown (20d):** Minor recent drawdown

Recent Signals:

Date	Close	State	Price State	OI State	CVD State
2025-07-09	0.29	Compression	UP	UP	UP
2025-07-10	0.29	Compression	UP	DOWN	UP
2025-07-11	0.30	Compression	UP	UP	DOWN
2025-07-12	0.30	Compression	UP	UP	DOWN
2025-07-13	0.30	Compression	UP	UP	UP

Daily Technical Analysis for TRXUSDT (180 periods)



Technical Interpretation

The current 8-state model signal is **Compression**. The asset is in a **strong uptrend**, with the current price (\$0.30) above both the 50-day SMA (\$0.28) and the 200-day SMA (\$0.25). Volatility is lower than average, indicating a period of consolidation. RSI is at **78.4**, indicating overbought conditions. MACD shows **bullish momentum**, with the MACD line above its signal line and a positive histogram.

Market State

Metric	Value
Daily Return (20d Avg)	-0.76%
Volatility State	Low
RSI	78.4 (Overbought)

Risk-Adjusted Returns

Metric	Value
Sharpe Ratio (20d)	9.18
Sortino Ratio	
- 20-day	11.10
- 60-day	1.83

Risk Metrics

Metric	20-day	60-day	120-day
CVaR	-1.50%	-3.36%	-
Omega Ratio	3.51	1.33	-
Max Drawdown	-1.50%	-9.98%	-9.98%

Appendix: Rule Definitions

Signal	Summary	Outlook
Bullish Trending	Price, Open Interest, and CVD are all signaling a strong upward trend. This is a high-conviction bullish signal suggesting broad market participation.	Bullish continuation is likely. Watch for OI or CVD to falter as an early warning sign of weakness.
Bullish Exhaustion	Price and Open Interest are rising, but aggressive selling (CVD down) is present. This suggests that while new money is entering, it's being met with significant selling pressure (absorption).	The upward move may be vulnerable to a pullback. If buyers overcome the selling (CVD turns up), the trend can resume. Otherwise, expect a stall or reversal.
Contrarian Bullish	Price and CVD are rising, but Open Interest is falling. This suggests the rally is fueled by short-sellers being forced to cover their positions, not by new organic longs.	This rally has limited fuel. Once short-covering is exhausted, the price is likely to stall. This is not a signal of a sustainable, healthy uptrend.
Bearish Trending	Price, Open Interest, and CVD are all signaling a strong downward trend. This is a high-conviction bearish signal.	The bearish trend is likely to continue. For a potential bottom, watch for CVD to flatten or turn up first.
Bearish Exhaustion	Price and Open Interest are falling, but CVD is rising. This suggests that while weak hands are capitulating, larger players may be quietly accumulating positions.	An early sign of potential bottoming. A sustained rise in CVD could signal a bullish reversal is forming.
Contrarian Bearish	Price is falling while Open Interest is rising and CVD is falling. This indicates that new short positions are being opened and are actively pushing the price down.	Strong bearish continuation is expected until new shorts stop entering the market.
Bullish Disagreement	Price is rising, but both Open Interest and CVD are falling. This suggests the up-move is weak, lacks broad participation, and is being sold into. It has the classic signs of a 'Bull Trap'.	A sharp reversal is common. This rally should be treated with extreme suspicion as it is unlikely to be sustained.
Bearish Disagreement	Price is falling, but both Open Interest and CVD are rising. This indicates a fierce battle where strong buyers are being met with even stronger overhead supply (sellers).	A significant spike in volatility is highly likely as the market is 'coiled'. The direction of the break is uncertain, but a decisive move is often imminent.
Noise (Bullish Bias)	The price move was not statistically significant (z-score < 0.5), indicating noise. However, the underlying OI and CVD flows show a bullish bias.	The market is consolidating without clear directional price intent. Wait for a breakout with a z-score > 0.5 before trusting the move.
Noise (Bearish Bias)	The price move was not statistically significant (z-score < 0.5), indicating noise. However, the underlying OI and CVD flows show a bearish bias.	The market is consolidating without clear directional price intent. Wait for a breakdown with a z-score < -0.5 before trusting the move.
Compression	The market is in a state of extremely low realized volatility. This indicates a 'compression' regime where the market is coiling and building energy.	Directional trades are low probability. Wait for volatility to expand, signaling the start of a new, more reliable move.
Undefined	The combination of price, OI, and CVD states does not match any predefined pattern.	Market conditions are unclear and choppy. It is best to wait for a recognizable signal to emerge.

Appendix: Volatility Metrics

Risk Metrics Glossary

- **Volatility State:** Current market volatility level (High/Normal/Low) based on return deviations.
- **RSI (Relative Strength Index):** Momentum oscillator measuring speed and change of price movements. Range: 0-100.
- **Sharpe Ratio:** Measures risk-adjusted returns relative to total volatility. Higher values indicate better risk-adjusted performance.
- **Sortino Ratio:** Measures risk-adjusted returns, focusing on downside volatility. Higher values indicate better risk-adjusted performance.
- **CVaR (Conditional Value at Risk):** Expected loss in the worst 5% of cases. More negative values indicate higher tail risk.
- **Omega Ratio:** Ratio of gains to losses above a threshold. Values > 1 indicate more gains than losses.
- **Max Drawdown:** Largest peak-to-trough decline. More negative values indicate larger price drops.

Interpretation Guidelines

Volatility State:

- **High:** Returns exceeding 2 standard deviations
- **Normal:** Returns within normal range
- **Low:** Returns below 0.5 standard deviations

RSI:

- **> 70:** Overbought conditions
- **< 30:** Oversold conditions
- **30-70:** Neutral conditions

Sharpe Ratio:

- **> 2:** Excellent risk-adjusted returns
- **1-2:** Good risk-adjusted returns
- **< 1:** Poor risk-adjusted returns

Sortino Ratio:

- **> 2:** Excellent risk-adjusted returns
- **1-2:** Good risk-adjusted returns
- **< 1:** Poor risk-adjusted returns

CVaR:

- **> -2%:** Low tail risk
- **-2% to -5%:** Moderate tail risk
- **< -5%:** High tail risk

Omega Ratio:

- **> 2:** Strong gain/loss ratio
- **1-2:** Moderate gain/loss ratio
- **< 1:** Weak gain/loss ratio

Max Drawdown:

- **> -5%:** Minor drawdown
- **-5% to -15%:** Moderate drawdown
- **< -15%:** Severe drawdown

Time Windows

Metrics are calculated over multiple time windows to provide different perspectives:

- **20 days:** Short-term view
- **60 days:** Medium-term view
- **120 days:** Long-term view (for Max Drawdown)